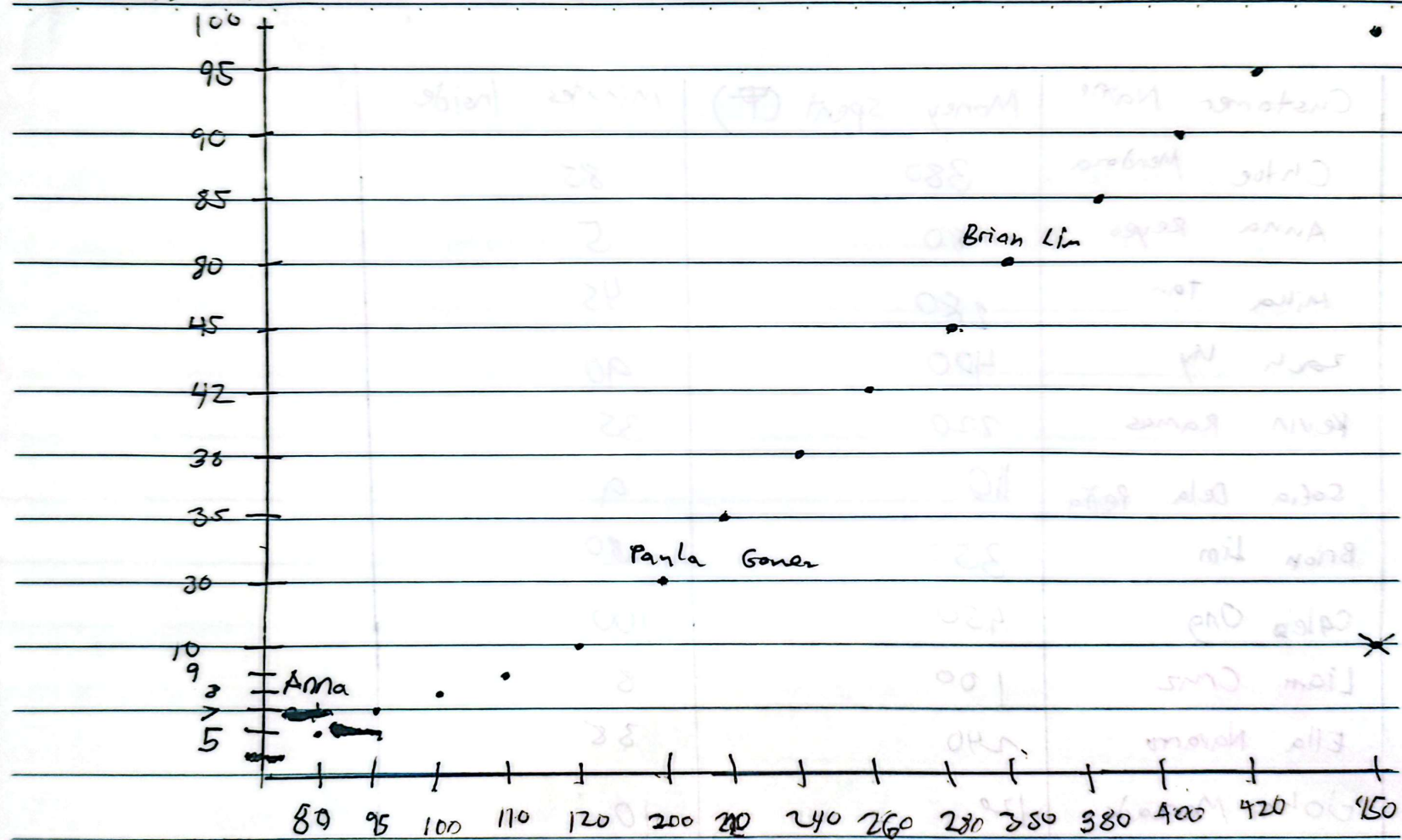


NO.:
DATE:

1. Plot the data in a scatter plot



NO.:

DATE:

	Customer Name	Money spent (\$)	Minutes inside	A d_1	B d_2	C d_3	Cluster
d_1	Chloe Mendoza	380	85	310.48 351.71	188.22 229.40	30.41	C
	Ana Reyes	80	5	0	122.58	280.22	A
	Mika Tan	290	45	203.96	81.39	78.26	C
	Zach Y	400	40	331.10	208.81	51.00	C
	Kerin Ramos	220	35	143.18	20.62	187.57	B
d_2	Sofia Dela Peña	110	9	30.27	92.45	250.28	A
	Brian Lim	350	80	280.22	158.11	0	C
	Caleb Ong	450	100	382	259.62	101.98	C
	Liam Cruz	100	8	20.22	102.39	280.16	A
	Ella Navarro	240	38	163.37	40.79	117.75	B
	John Mercado	120	10	40.31	82.46	240.42	A
	Jared Flores	260	42	183.76	61.19	97.69	B
d_3	Mark Santos	95	7	15.13	107.49	265.24	A
	Paula Gomez	200	30	122.58	0	158.11	B
	Hannah Rague	420	95	351.71	229.40	71.59	C

Hannah Rague (420, 95)

Paula Gomez

Mark Santos

$$d_1 = \sqrt{(420-80)^2 + (95-5)^2}$$

$$= \sqrt{123,700}$$

$$= 351.71$$

$$d_1 = \sqrt{(200-80)^2 + (30-5)^2}$$

$$= \sqrt{15025}$$

$$= 122.58$$

$$d_1 = \sqrt{(95-10)^2 + (7-5)^2}$$

$$= \sqrt{229}$$

$$= 15.13$$

$$d_2 = \sqrt{(420-200)^2 + (95-30)^2}$$

$$= \sqrt{52,625}$$

$$= 229.40$$

$$d_2 = \sqrt{(200-200)^2 + (30-30)^2}$$

$$= \sqrt{0}$$

$$= 0$$

$$d_2 = \sqrt{(95-200)^2 + (7-30)^2}$$

$$= \sqrt{11554}$$

$$= 107.49$$

$$d_3 = \sqrt{(420-350)^2 + (95-80)^2}$$

$$= \sqrt{5125}$$

$$= 71.59$$

$$d_3 = \sqrt{(200-350)^2 + (30-80)^2}$$

$$= \sqrt{25000}$$

$$= 158.11$$

$$d_3 = \sqrt{(95-350)^2 + (7-80)^2}$$

$$= \sqrt{70854}$$

$$= 265.24$$

Jared Flores

$$d_1 = \sqrt{(260-80)^2 + (42-5)^2}$$

$$= \sqrt{37769}$$

$$= 183.76$$

$$d_2 = \sqrt{(260-200)^2 + (42-30)^2}$$

$$= \sqrt{3744}$$

$$= 61.19$$

$$d_3 = \sqrt{(260-350)^2 + (42-80)^2}$$

$$= \sqrt{9544}$$

$$= 97.69$$

John Mercado	Ella Navano	Liam Cruz
$d_1 = \sqrt{(120-80)^2 + (10-5)^2}$ $= \sqrt{1625}$ $= 40.31$	$d_1 = \sqrt{(240-80)^2 + (38-5)^2}$ $= \sqrt{26689}$ $= 163.37$	$d_1 = \sqrt{(100-80)^2 + (8-5)^2}$ $= \sqrt{409}$ $= 20.22$
$d_2 = \sqrt{(120-200)^2 + (10-30)^2}$ $= \sqrt{5800}$ $= 82.46$	$d_2 = \sqrt{(240-200)^2 + (38-30)^2}$ $= \sqrt{1664}$ $= 40.79$	$d_2 = \sqrt{(100-200)^2 + (8-30)^2}$ $= \sqrt{10484}$ $= 102.39$
$d_3 = \sqrt{(120-350)^2 + (10-80)^2}$ $= \sqrt{57802}$ $= 240.42$	$d_3 = \sqrt{(240-350)^2 + (38-80)^2}$ $= \sqrt{13864}$ $= 117.75$	$d_3 = \sqrt{(100-350)^2 + (8-80)^2}$ $= \sqrt{57684}$ $= 260.16$
Caleb Ong	Brian Lim	Sofia De la Peña
$d_1 = \sqrt{(450-80)^2 + (800-5)^2}$ $= \sqrt{145925}$ $= 382.00$	$d_1 = \sqrt{(350-80)^2 + (800-5)^2}$ $= \sqrt{78525}$ $= 280.22$	$d_1 = \sqrt{(110-80)^2 + (9-5)^2}$ $= \sqrt{916}$ $= 30.27$
$d_2 = \sqrt{(450-200)^2 + (800-30)^2}$ $= \sqrt{67400}$ $= 259.62$	$d_2 = \sqrt{(350-200)^2 + (800-30)^2}$ $= \sqrt{25000}$ $= 158.11$	$d_2 = \sqrt{(110-200)^2 + (9-30)^2}$ $= \sqrt{8541}$ $= 92.45$
$d_3 = \sqrt{(450-350)^2 + (800-80)^2}$ $= \sqrt{10400}$ $= 101.98$	$d_3 = \sqrt{(350-350)^2 + (800-80)^2}$ $= \sqrt{0}$ $= 0$	$d_3 = \sqrt{(110-350)^2 + (9-80)^2}$ $= \sqrt{62641}$ $= 250.28$
Kevin Ramos	Zach Uy	Mika Tan
$d_1 = \sqrt{(620-80)^2 + (35-5)^2}$ $= \sqrt{20500}$ $= 143.18$	$d_1 = \sqrt{(400-80)^2 + (90-5)^2}$ $= \sqrt{109625}$ $= 331.10$	$d_1 = \sqrt{(280-80)^2 + (45-5)^2}$ $= \sqrt{41600}$ $= 203.96$
$d_2 = \sqrt{(620-200)^2 + (35-30)^2}$ $= \sqrt{425}$ $= 20.62$	$d_2 = \sqrt{(400-200)^2 + (90-30)^2}$ $= \sqrt{43600}$ $= 208.81$	$d_2 = \sqrt{(280-200)^2 + (45-30)^2}$ $= \sqrt{6625}$ $= 81.39$
$d_3 = \sqrt{(620-350)^2 + (35-80)^2}$ $= \sqrt{18925}$ $= 137.57$	$d_3 = \sqrt{(400-350)^2 + (90-80)^2}$ $= \sqrt{2600}$ $= 51.00$	$d_3 = \sqrt{(280-350)^2 + (45-80)^2}$ $= \sqrt{6125}$ $= 78.26$

$$q_1 = \frac{(95 + 100 + 110 + 80)}{5} = \frac{505}{5} = 101$$

New $d_1 = (101, 7.8)$

$$b_1 = \frac{(7 + 10 + 8 + 9 + 5)}{5} = \frac{39}{5} = 7.8$$

$$q_2 = \frac{(200 + 260 + 243 + 220)}{4} = \frac{923}{4} = 230.75$$

New $d_2 = (230.75, 36.25)$

$$b_2 = \frac{(30 + 42 + 38 + 35)}{4} = \frac{145}{4} = 36.25$$

$$q_3 = \frac{(420 + 450 + 350 + 400 + 280 + 380)}{6} = \frac{2280}{6} = 380$$

New $d_3 = (380, 82.5)$

$$b_3 = \frac{(95 + 100 + 80 + 90 + 45 + 85)}{6} = \frac{165}{2} = 82.5$$

Customer Name	Money Spent (\$)	Minutes Inside	A d_1	B d_2	C d_3	Cluster
Chloe Mendoza	380	85	289.48	157.01	2.5	C
Ama Reyes	80	5	21.19	153.95	309.85	A
Mika Tan	280	45	182.82	50.02	106.80	B
Zach Uy	400	90	310.10	177.58	21.36	C
Kerith Ramos	220	35	122.07	10.33	166.90	B
Sofia Dela Peña	110	9	9.08	123.79	274.83	A
Brian Lim	350	80	259.26	127.02	30.10	C
Caleb Ong	450	100	360.97	228.33	72.154	C
Liam Cruz	100	8	1.02	133.77	289.71	A
Ella Navarro	740	38	142.24	9.41	146.90	B
John Mercado	120	10	19.12	113.22	269.92	A
Wared Flores	260	42	162.64	29.80	126.65	B
Mark Santos	95	7	6.05	138.27	294.23	A
Anita Gomez	200	30	101.46	31.38	187.5	B
Hannah Roque	420	95	330.70	198.16	41.91	C

Anna Reyes

$$d_1 = \sqrt{(80 - 80)^2 + (5 - 5)^2}$$

$$= \sqrt{0}$$

$$= 0$$

Chloe Mendoza

$$d_1 = \sqrt{(380 - 80)^2 + (85 - 5)^2}$$

$$= \sqrt{96400}$$

$$= 310.48$$

$$d_2 = \sqrt{(80 - 200)^2 + (5 - 30)^2}$$

$$= \sqrt{18025}$$

$$= 122.58$$

$$d_2 = \sqrt{(380 - 200)^2 + (85 - 30)^2}$$

$$= \sqrt{35425}$$

$$= 188.22$$

$$d_3 = \sqrt{(80 - 350)^2 + (5 - 80)^2}$$

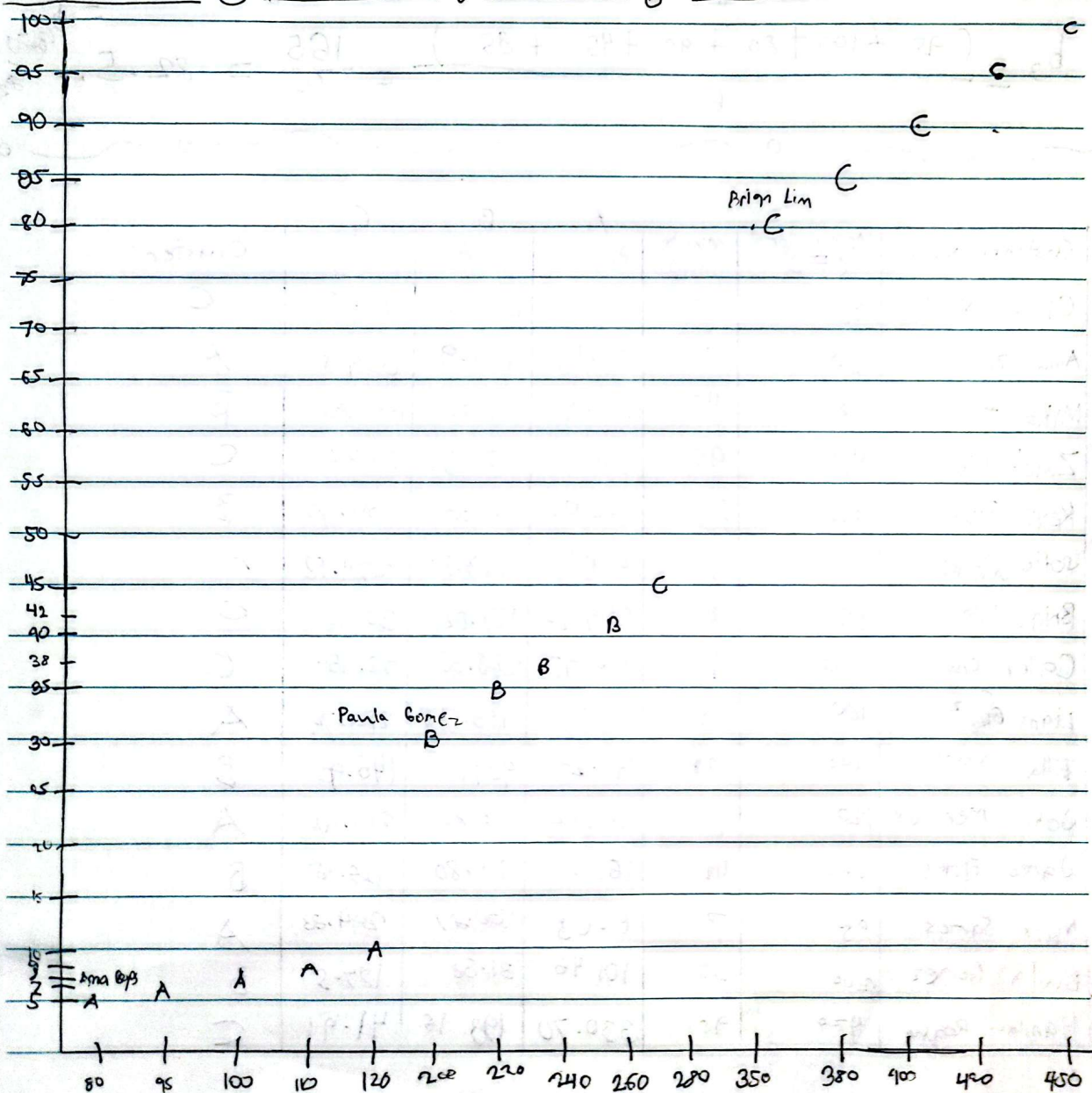
$$= \sqrt{78525}$$

$$= 280.22$$

$$d_3 = \sqrt{(380 - 350)^2 + (85 - 80)^2}$$

$$= \sqrt{925}$$

$$= 30.41$$



Chloe Mendoza

$$d_1 = \sqrt{(380-101)^2 + (85-7.8)^2} \quad d_2 = \sqrt{(380-230.75)^2 + (85-36.25)^2} \quad d_3 = \sqrt{(380-380)^2 + (85-82.5)^2}$$

$$= \sqrt{83800.84} \quad = \sqrt{24652.125} \quad = \sqrt{6.25}$$

$$= 289.48 \quad = 157.01 \quad = 2.5$$

Ama Reyes

$$d_1 = \sqrt{(380-101)^2 + (85-7.8)^2} \quad d_2 = \sqrt{(380-230.75)^2 + (85-36.25)^2} \quad d_3 = \sqrt{(380-380)^2 + (85-82.5)^2}$$

$$= \sqrt{448.81} \quad = \sqrt{24652.125} \quad = \sqrt{96006.25}$$

$$= 21.19 \quad = 153.95 \quad = 309.85$$

Mika Tan

$$d_1 = \sqrt{(280-101)^2 + (45-7.8)^2} \quad d_2 = \sqrt{(280-230.75)^2 + (45-36.25)^2} \quad d_3 = \sqrt{(280-380)^2 + (45-82.5)^2}$$

$$= \sqrt{33424.84} \quad = \sqrt{2502.125} \quad = \sqrt{11406.25}$$

$$= 182.82 \quad = 50.02 \quad = 106.80$$

Zach Uy

$$d_1 = \sqrt{(400-101)^2 + (90-7.8)^2} \quad d_2 = \sqrt{(400-230.75)^2 + (90-36.25)^2} \quad d_3 = \sqrt{(400-380)^2 + (90-82.5)^2}$$

$$= \sqrt{95157.84} \quad = \sqrt{31534.625} \quad = \sqrt{456.25}$$

$$= 310.10 \quad = 177.58 \quad = 21.36$$

Kevin Ramos

$$d_1 = \sqrt{(400-101)^2 + (90-7.8)^2} \quad d_2 = \sqrt{(400-230.75)^2 + (90-36.25)^2} \quad d_3 = \sqrt{(400-380)^2 + (90-82.5)^2}$$

$$= \sqrt{14900.84} \quad = \sqrt{106.625} \quad = \sqrt{27.856.25}$$

$$= 122.07 \quad = 10.33 \quad = 166.90$$

Sofia Dela Peña

$$d_1 = \sqrt{(400-101)^2 + (90-7.8)^2} \quad d_2 = \sqrt{(400-230.75)^2 + (90-36.25)^2} \quad d_3 = \sqrt{(400-380)^2 + (90-82.5)^2}$$

$$= \sqrt{82.44} \quad = \sqrt{15323.125} \quad = \sqrt{78302.25}$$

$$= 9.08 \quad = 123.79 \quad = 279.83$$

Brian Lim

$$d_1 = \sqrt{(350-101)^2 + (80-7.8)^2} \quad d_2 = \sqrt{(350-230.75)^2 + (80-36.25)^2} \quad d_3 = \sqrt{(350-380)^2 + (80-82.5)^2}$$

$$= \sqrt{67213.84} \quad = \sqrt{16134.625} \quad = \sqrt{906.25}$$

$$= 259.26 \quad = 127.02 \quad = 30.10$$

Caleb Ong

$$d_1 = \sqrt{(450-101)^2 + (100-7.8)^2} \quad d_2 = \sqrt{(450-230.75)^2 + (100-36.25)^2} \quad d_3 = \sqrt{(450-380)^2 + (100-82.5)^2}$$

$$= \sqrt{130301.84} \quad = \sqrt{52134.625} \quad = \sqrt{5206.25}$$

$$= 360.97 \quad = 228.33 \quad = 72.154$$

Liam Cruz

$$d_1 = \sqrt{(100 - 101)^2 + (8 - 7.8)^2} = \sqrt{(100 - 230.75)^2 + (8 - 36.25)^2} = \sqrt{(100 - 380)^2 + (8 - 82.5)^2}$$

$$= \sqrt{1.04} \quad = \sqrt{17893.625} \quad = \sqrt{83950.25}$$

$$= 1.02 \quad = 133.77 \quad = 289.74$$

Ella Navarro

$$d_1 = \sqrt{(240 - 101)^2 + (38 - 7.8)^2} = \sqrt{(240 - 230.75)^2 + (38 - 36.25)^2} = \sqrt{(240 - 380)^2 + (38 - 82.5)^2}$$

$$= \sqrt{20233.04} \quad = \sqrt{88.625} \quad = \sqrt{21580.25}$$

$$= 142.24 \quad = 9.41 \quad = 146.90$$

John Mercado

$$d_1 = \sqrt{(120 - 101)^2 + (10 - 7.8)^2} = \sqrt{(120 - 230.75)^2 + (10 - 36.25)^2} = \sqrt{(120 - 380)^2 + (10 - 82.5)^2}$$

$$= \sqrt{365.84} \quad = \sqrt{12954.625} \quad = \sqrt{72856.25}$$

$$= 19.12 \quad = 113.82 \quad = 269.92$$

Jared Flores

$$d_1 = \sqrt{(256 - 101)^2 + (42 - 7.8)^2} = \sqrt{(256 - 230.75)^2 + (42 - 36.25)^2} = \sqrt{(256 - 380)^2 + (42 - 82.5)^2}$$

$$= \sqrt{26450.64} \quad = \sqrt{888.625} \quad = \sqrt{16040.25}$$

$$= 162.64 \quad = 29.80 \quad = 126.65$$

Mark Santos

$$d_1 = \sqrt{(200 - 101)^2 + (20 - 7.8)^2} = \sqrt{(200 - 230.75)^2 + (20 - 36.25)^2} = \sqrt{(200 - 380)^2 + (20 - 82.5)^2}$$

$$= \sqrt{35.64} \quad = \sqrt{19283.625} \quad = \sqrt{86925.25}$$

$$= 6.05 \quad = 138.87 \quad = 294.83$$

Paula Gomez

$$d_1 = \sqrt{(200 - 101)^2 + (30 - 7.8)^2} = \sqrt{(200 - 230.75)^2 + (30 - 36.25)^2} = \sqrt{(200 - 380)^2 + (30 - 82.5)^2}$$

$$= \sqrt{10293.84} \quad = \sqrt{984.0625} \quad = \sqrt{35156.25}$$

$$= 101.46 \quad = 31.38 \quad = 187.5$$

Hannah Roque

$$d_1 = \sqrt{(420 - 101)^2 + (95 - 7.8)^2} = \sqrt{(420 - 230.75)^2 + (95 - 36.25)^2} = \sqrt{(420 - 380)^2 + (95 - 82.5)^2}$$

$$= \sqrt{109364.84} \quad = \sqrt{39267.125} \quad = \sqrt{1756.25}$$

$$= 330.70 \quad = 198.16 \quad = 41.91$$

